

CLINICAL AND POPULATION SCIENCES



MAD-MT Score: A Tool to Optimize Patient Selection for Mechanical Thrombectomy in Distal Vessel Occlusions

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BACKGROUND: Endovascular thrombectomy (EVT) for distal and medium vessel occlusion stroke remains uncertain. We aimed to develop a medium and distal mechanical thrombectomy score integrating clinical need and procedural risk to guide patient selection.

METHODS: This retrospective cohort analysis used an international distal and medium vessel occlusion stroke registry spanning the study period of 2017 to 2023. Patients with acute distal and medium vessel occlusion stroke who received medical management (MM) or EVT across 37 stroke centers were included, and those with baseline modified Rankin Scale score of ≥ 3 or those with missing covariable data were excluded. Multivariable logistic regression identified predictors of poor functional outcome (modified Rankin Scale score > 2 at 90 days) in the MM cohort and predictors of EVT failure/complications in the EVT cohort. Predictors of poor outcome on MM were assigned positive weights (clinical need); predictors of EVT failure/harm were assigned negative weights (procedural risk). The medium and distal mechanical thrombectomy score summed these weighted points. Interaction analysis assessed the heterogeneity of EVT effect by score.

RESULTS: A total of 1217 patients were identified, and 1007 were included (EVT: 822, MM: 185; median age 73 years; 41% female). Higher National Institutes of Health Stroke Scale score (+1 per point) and lack of intravenous thrombolysis (+7) predicted poor MM outcomes. In the EVT cohort, older age (−1 per 15 years above 25), absence of hypertension (−2), and absence of atrial fibrillation (−2) predicted failure/complications. The medium and distal mechanical thrombectomy score (range −8 to 49) significantly modified EVT effect versus MM ($P_{\text{interaction}} = 0.048$). In high-score patients (≥ 15 ; $n = 293$), EVT yielded a better 90-day modified Rankin Scale score than MM (median, 3 versus 4; $P = 0.009$). Conversely, in low-score patients (< 15 ; $n = 710$), EVT yielded a worse modified Rankin Scale score (median, 2 versus 1; $P = 0.014$).

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CONCLUSIONS: The medium and distal mechanical thrombectomy score is a pragmatic tool that identifies patients with distal and medium vessel occlusion most likely to benefit from EVT while minimizing risk, supporting patient-centered decisions and future trial design.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: angiography ■ ischemic stroke ■ prognosis ■ subarachnoid hemorrhage ■ thrombectomy

Nonstandard Abbreviations and Acronyms

DMVO	distal and medium vessel occlusion
EVT	endovascular thrombectomy
ICH	intracranial hemorrhage
IVT	intravenous thrombolysis
MAD-MT	medium and distal mechanical thrombectomy
MM	medical management
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
OR	odds ratio
RCS	restricted cubic spline

While endovascular thrombectomy (EVT) is a well-established and highly effective treatment for large vessel occlusion strokes, its role in distal and medium vessel occlusion (DMVO) strokes remains uncertain. Recent randomized controlled trials, including DISTAL (Endovascular Treatment for Medium or Distal Vessel Occlusion Stroke)¹ and ESCAPE-MeVO (Endovascular Treatment to Improve Outcomes for Medium Vessel Occlusions),² have reported neutral outcomes when comparing DMVO-EVT to medical management (MM), raising questions about its relative benefit in this population. Unlike large vessel occlusions, DMVO strokes often present with milder, yet highly variable, clinical deficits. In DISTAL and ESCAPE-MeVO, 55% of medically managed patients with DMVO were functionally independent at 90 days, whereas 45% were not.^{1,2} While these findings confirm that many patients with DMVO still experience debilitating neurological deficits, they also suggest that many patients do well with MM alone, highlighting the large variability in patient prognosis and, by extension, a lack of ubiquitous need for more aggressive treatments, such as EVT.

Beyond the variable need for EVT among patients with DMVO, current evidence also suggests that DMVO-EVT may carry increased procedural risks compared with large vessel occlusion-EVT, including vessel perforation and subarachnoid hemorrhage.^{3–7} Challenges such as smaller vessel caliber, more distal clot locations, and complex endovascular navigation may contribute to

higher rates of procedural complications and lower rates of technical success. Given the typically milder baseline deficits in patients with DMVO, the risks of procedural harm may outweigh potential benefits in many cases.

Together, these complexities may have partially explained the neutral results of recent trials, and they highlight the need for a more refined, patient-centered decision-making framework for patients with DMVO: one that balances clinical need (likelihood of poor functional outcomes without EVT) and odds of procedural success. While many tools exist to predict post-EVT outcomes using pre-EVT variables,^{8–12} there are currently no established methods to predict EVT’s benefit compared with MM to guide optimal EVT selection for patients with DMVO. In this multicenter retrospective study, we used a multivariable, risk-based, predictive approach to develop a holistic medium and distal mechanical thrombectomy (MAD-MT) score to identify patients with DMVO most likely to derive meaningful benefit from EVT.

METHODS

Database Characteristics

The Multicenter Analysis of Primary Distal Medium Vessel Occlusions: Effect of Mechanical Thrombectomy (MAD-MT) registry is a multinational, retrospective-prospective cohort study involving 37 academic centers across North America, Asia, and Europe. The primary cohort was previously published, and this is a secondary and novel analysis of this previously published cohort.⁴ Data collection spanned from September 2017 to July 2023, incorporating both prospective registry entries and retrospective case reviews. This report adheres to STROBE guidelines (Strengthening the Reporting of Observational Studies in Epidemiology), and ethical approval was obtained from all participating institutions, with waived informed consent due to the retrospective nature of the analysis. All clinical, imaging, and outcome variables were site-collected without central adjudication or use of an imaging core laboratory. Consecutive patient inclusion was verified through site-level attestation.

Patient Population

Consecutive patients who are functionally independent at baseline with acute ischemic stroke due to DMVO, defined as occlusions in the M2/M3/M4, A1/A2/A3, or P1/P2/P3 segments managed with EVT or MM, were included. DMVOs were confirmed via angiography. Patients functionally dependent at baseline (baseline modified Rankin Scale [mRS] score >2) were excluded. Those with missing laterality (left versus right-sided

stroke), missing admission National Institutes of Health Stroke Scale (NIHSS) information, primary treatment with intra-arterial thrombolysis, missing information on intravenous thrombolysis (IVT), or vascular occlusion other than DMVO criteria stated above were also excluded.

Clinical Variables

Baseline demographic and clinical variables were systematically collected, including age, sex, and smoking status, as well as medical history such as hypertension, diabetes, hypercholesterolemia, and atrial fibrillation. Stroke severity was assessed using the NIHSS score at admission. Prestroke mRS¹³ score, site of occlusion, and stroke laterality were documented. Procedural variables encompassed treatment modality (EVT and IVT). Time from last known normal to arteriotomy for the EVT group was also captured and categorized into early (<4.5 hours), late (≥4.5 hours), wake-up stroke, or unknown. Final modified treatment in cerebral ischemia (modified Thrombolysis in Cerebral Infarction¹⁴) score was recorded, as were occurrences of intracranial hemorrhages (ICHs, graded by thrombolytic therapy with intravenous alteplase in acute ischemic stroke, ECASS-II classification), with symptomatic ICH defined as parenchymal hematoma or subarachnoid hemorrhage associated with a 4-point or more worsening of NIHSS score 24 hours after hospital admission.¹⁵ Finally, procedural complications were also collected, including vessel perforation, embolism to new territory, vessel dissection, and intraprocedural vasospasm. Of note, imaging markers such as Alberta Stroke Program Early Computed Tomography Score were excluded from this analysis as Alberta Stroke Program Early Computed Tomography Score is only applicable to infarcts within the middle cerebral artery territory, and its reliability is unproven in the setting of DMVOs.¹⁶ Due to variability in institutional practices across participating sites, data on collateral status, perfusion metrics, and magnetic resonance imaging markers were not consistently available or collected; thus, these markers were also not included in this study. Clinical, imaging, and outcome variables were site-collected without central adjudication or use of an imaging core laboratory.

Study Design and Outcome Variables

Our analysis used established methods for evaluating heterogeneity of treatment effects using a multivariable, predictive, risk-based approach adapted from Rothwell and Warlow.¹⁷ The study cohort was stratified into 2 treatment arms: MM and EVT.

In the MM cohort, we identified independent predictors of 90-day functional dependence (mRS score >2). Positive weights were assigned to these predictors based on their effect sizes, with higher scores reflecting greater clinical need for EVT. Conversely, within the EVT cohort, we evaluated predictors of procedural complications or unsuccessful reperfusion, defined as modified Thrombolysis in Cerebral Infarction score <2b, symptomatic ICH, vessel perforation, embolization to new territory, symptomatic dissection (NIHSS score worsening ≥4), or symptomatic vasospasm (NIHSS score worsening ≥4). These factors received negative weights proportional to their effect sizes, indicating increased procedural risk. The aggregated sum of these weighted predictors was defined as the MAD-MT score.

We then assessed treatment outcomes across the entire cohort using multiple end points: ordinal analysis of 90-day mRS score, functional independence (mRS score, 0–2), severe

disability or death (mRS score, 5–6), symptomatic ICH, early neurological improvement (NIHSS score decrease ≥4), and early neurological deterioration (NIHSS score increase ≥4). Interaction and subgroup analyses incorporating the MAD-MT score were performed to evaluate heterogeneity of treatment effects and the score's ability to identify patients with DMVO stroke most likely to benefit from EVT.

Statistical Methods

Descriptive statistics for categorical variables were reported as frequencies and percentages (%), while continuous variables were summarized as medians with quartiles (Q1–Q3). Bivariate comparisons between treatment groups were conducted using the Pearson χ^2 tests for categorical variables and the Wilcoxon rank-sum tests for continuous variables.

To identify independent predictors of poor functional outcomes (mRS score >2 at 90 days) in the MM cohort and procedural failure or complications in the EVT cohort, we constructed multivariable logistic regression models incorporating all baseline clinical, imaging, and treatment-related variables. Backward elimination using the Akaike information criterion was performed.¹⁸ In brief, starting with the full model containing all clinically relevant baseline characteristics, variables were sequentially removed based on the Akaike information criterion, with a penalty term ($k=2$), until no further improvement in model fit was achieved. Only cases with complete data were included in these regression analyses. To confirm the linearity of continuous predictors (eg, patient age and NIHSS score) in their associations with the respective outcomes, restricted cubic splines (RCS) with 4 knots were used. For each final stepwise-selected multivariable model, the linear term of the continuous predictor was replaced with an RCS transformation, and the model was reestimated using penalized maximum likelihood logistic regression. An ANOVA was then performed on the RCS model, which decomposes each spline term into its linear and nonlinear components. A statistically significant P value for the nonlinear component indicates departure from a log-linear relationship between the predictor and the log-odds of the outcome.

In the final models, variables with $P<0.05$ were assigned a score value proportional to their effect size (per 0.25-unit increment of log-odds, rounded to the nearest integer). Predictors of poor outcomes in the MM group received positive weights (indicating greater potential benefit from EVT), while predictors of procedural failure or complications in the EVT group were assigned negative weights (reflecting higher procedural risk). The MAD-MT score was calculated for each patient as the arithmetic sum of these weighted predictors.

In the overall cohort, outcomes between EVT and MM were compared using χ^2 tests (for binary end points) or Wilcoxon rank-sum tests (for ordinal/continuous variables). To evaluate heterogeneity of treatment effects, we performed interaction analyses using logistic regression (for binary outcomes) and ordinal regression (for mRS score shift analysis), incorporating the MAD-MT score as an effect modifier. To ensure that integer rounding of regression coefficients did not materially alter MAD-MT score's ability to identify heterogeneity of treatment effects, interaction analyses were also conducted on unrounded MAD-MT scores. Additional sensitivity analyses were also conducted to assess the MAD-MT score's ability to characterize

treatment heterogeneity specifically among M2 and non-M2 occlusion patients using subgroup analyses of 2-way interaction terms and 3-way interaction terms. Marginal probability plots were generated to visualize the relationship between the MAD-MT score and treatment benefit. Based on these plots, an optimal MAD-MT score threshold was visually identified to dichotomize patients into high- versus low-score subgroups, maximizing discrimination of treatment response. This threshold was used to dichotomize the study cohort, and outcomes of EVT versus MM were assessed in the high MAD-MT score and low MAD-MT score groups. All analyses were conducted using R statistical software (version 4.3.1). Outcome comparisons of EVT versus MM stratified by MAD-MT score were not additionally adjusted for confounders, as the MAD-MT score already contains key predictors of patient outcomes to account for confounding bias, and additional inclusion of covariates would introduce excessive collinearity. Statistical significance was defined as a 2-tailed $P < 0.05$. Missing data were not imputed; all analyses were performed on complete cases only.

RESULTS

Study Cohort

Among a total of 1217 patients in the full registry, 1007 met study inclusion/exclusion criteria (Figure 1). Overall, 822 patients received EVT, and 185 received MM only. Patient characteristics are detailed in Table 1, largely comparable to DISTAL and ESCAPE-MeVO except for a higher median NIHSS score among the EVT group (10 in our study, compared with 6 and 8 in DISTAL and ESCAPE-MeVO, respectively), likely due to our study's lack of differentiation between dominant and nondominant M2 occlusions, thereby enriching our EVT cohort with more symptomatic patients.^{1,2} In terms of data missingness within our study cohort, 4 patients (0.4%, 2 in EVT and 2 in MM) had missing age information, and 81 patients (8.0%, 70 in EVT and 11 in MM) had missing baseline mRS scores. There were no missing data for all other study variables.

MM Group and Clinical Need for EVT

Among the 185 patients who underwent MM, 64 (34.6%) had poor outcomes, defined as functional dependence (mRS score >2) at 90 days. Compared with MM patients with good outcomes, those with poor outcomes were significantly older (79 versus 68 years; $P=0.002$) and had higher NIHSS scores (median, 13 versus 5; $P<0.001$). Those with poor outcomes also had higher rates of atrial fibrillation (45% versus 22%; $P=0.002$) and lower rates of IVT treatment (39% versus 60%; $P=0.009$; Table 1).

EVT Group and Failure or Complications

Among 822 EVT patients, 99 (12%) did not have successful recanalization (modified Thrombolysis in Cerebral Infarction $<2b$), 39 (4.7%) had vessel perforation,

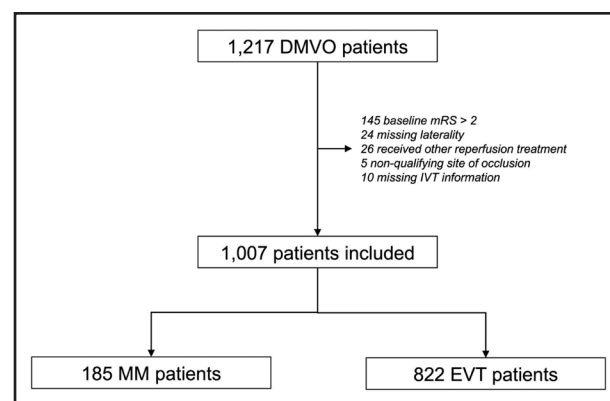


Figure 1. Participant flowchart.

DMVO indicates distal or medium vessel occlusion; EVT, endovascular thrombectomy; IVT, intravenous thrombolysis; MM, medical management; and mRS, modified Rankin Scale.

23 (2.8%) had embolism to new territory, 18 (2.2%) had symptomatic ICH, 2 (0.2%) had symptomatic dissection, and 1 (0.1%) had intraprocedural thrombectomy device fracture. Together, 157 (19.7%) experienced unsuccessful EVT reperfusion or major complications. Compared with patients who underwent successful and uncomplicated EVT, those who had unsuccessful or complicated EVT had significantly lower rates of atrial fibrillation (27% versus 37%; $P=0.015$), lower rates of hypertension (66% versus 76%; $P=0.016$), and higher NIHSS scores (median, 12 versus 10; $P=0.047$; Table 1).

Creating the MAD-MT Score

First, to predict clinical need for EVT, we used a multivariable regression with a backward selection method to predict poor outcomes among the MM group. Here, only NIHSS score (odds ratio [OR], 1.24 [95% CI, 1.16–1.34] per point; $P<0.001$) and lack of IVT (OR, 5.33 [95% CI, 2.24–14.0]; $P<0.001$) were significantly associated with higher odds of poor outcomes (and, thus, clinical need) in the final model. ANOVA testing on the RCS model, decomposing NIHSS score into linear and nonlinear components, confirmed that there was no significant non-linear relationship between NIHSS score and its association with poor outcomes in the MM group ($P=0.10$). Based on their effect sizes, each NIHSS score point was assigned a +1 value, and lack of IVT was assigned a +7 value toward the MAD-MT score (Table 2).

Next, to predict procedural failure, we performed a separate multivariable regression with backward selection in the EVT group. Here, older age (OR, 1.35 [95% CI, 1.07–1.70] for every 15 years; $P=0.012$), lack of hypertension (OR, 1.85 [95% CI, 1.20–2.78]; $P=0.004$), and lack of atrial fibrillation (OR, 1.61 [95% CI, 1.08–2.43]; $P=0.024$) were identified as significant predictors of procedural failure. ANOVA testing on the RCS model, decomposing patient age into linear and nonlinear components, confirmed that there was no significant

Table 1. Patient Characteristics

Characteristic, n (%) or median (Q1–Q3)	MM				EVT			
	All MM patients (n=185)	Good outcomes (mRS score ≤2; n=121)	Poor outcomes (mRS score >2; n=64)	P value*	All EVT patients (n=822)	EVT success (n=657)	EVT failure (n=165)	P value*
Female sex	75 (41%)	48 (40%)	27 (42%)	0.86	420 (51%)	331 (50%)	89 (54%)	0.47
Age, y	73 (61–82)	68 (59–80)	79 (67–83)	0.002	74 (62–81)	74 (62–81)	75 (65–82)	0.21
Comorbidities								
Atrial fibrillation	56 (30%)	27 (22%)	29 (45%)	0.002	288 (35%)	244 (37%)	44 (27%)	0.015
Hypertension	127 (69%)	79 (65%)	48 (75%)	0.24	606 (74%)	497 (76%)	109 (66%)	0.016
Hyperlipidemia	95 (51%)	60 (50%)	35 (55%)	0.61	294 (36%)	245 (37%)	49 (30%)	0.084
Diabetes	53 (29%)	34 (28%)	19 (30%)	0.96	180 (22%)	148 (23%)	32 (19%)	0.44
Smoking	26 (14%)	20 (17%)	6 (9.4%)	0.27	97 (12%)	80 (12%)	17 (10%)	0.59
Baseline mRS score								
0 (asymptomatic)	133 (76%)	92 (79%)	41 (72%)	0.24	516 (69%)	413 (68%)	103 (72%)	0.38
1 (symptomatic)	29 (17%)	20 (17%)	9 (16%)		138 (18%)	113 (19%)	25 (17%)	
2 (disabled but independent)	12 (6.9%)	5 (4.3%)	7 (12%)		98 (13%)	82 (13%)	16 (11%)	
Stroke onset to intervention								
Early (<4.5 h)	25 (14%)	18 (15%)	7 (11%)	0.27	344 (42%)	278 (42%)	66 (40%)	0.5
Late (≥4.5 h)	51 (28%)	38 (31%)	13 (20%)		192 (23%)	148 (23%)	44 (27%)	
Wake-up stroke	22 (12%)	13 (11%)	9 (14%)		58 (7.1%)	44 (6.7%)	14 (8.5%)	
Unknown	87 (47%)	52 (43%)	35 (55%)		228 (28%)	187 (28%)	41 (25%)	
National Institutes of Health Stroke Scale score	7 (4–13)	5 (3–8)	13 (9–20)	<0.001	10 (6–16)	10 (6–16)	12 (6–18)	0.047
Occlusion site								
M2	92 (50%)	53 (44%)	39 (61%)	0.09	628 (76%)	505 (77%)	123 (75%)	0.32
M3/M4	27 (15%)	22 (18%)	5 (7.8%)		112 (14%)	89 (14%)	23 (14%)	
P1	9 (4.9%)	8 (6.6%)	1 (1.6%)		16 (1.9%)	12 (1.8%)	4 (2.4%)	
P2/P3	43 (23%)	28 (23%)	15 (23%)		32 (3.9%)	28 (4.3%)	4 (2.4%)	
Anterior cerebral artery	14 (7.6%)	10 (8.3%)	4 (6.3%)		34 (4.1%)	23 (3.5%)	11 (6.7%)	
Left side	103 (56%)	69 (57%)	34 (53%)	0.72	418 (51%)	324 (49%)	94 (57%)	0.095
IVT	98 (53%)	73 (60%)	25 (39%)	0.009	407 (50%)	326 (50%)	81 (49%)	0.97
Hemorrhagic transformation								
None	166 (90%)	108 (89%)	58 (91%)	0.38	564 (69%)	472 (72%)	92 (56%)	<0.001
HI1	12 (6.5%)	10 (8.3%)	2 (3.1%)		146 (18%)	109 (17%)	37 (22%)	
HI2	2 (1.1%)	1 (0.8%)	1 (1.6%)		10 (1.2%)	9 (1.4%)	1 (0.6%)	
PH1	4 (2.2%)	2 (1.7%)	2 (3.1%)		20 (2.4%)	12 (1.8%)	8 (4.8%)	
PH2	1 (0.5%)	0 (0%)	1 (1.6%)		15 (1.8%)	10 (1.5%)	5 (3.0%)	
Subarachnoid hemorrhage	0 (0%)	0 (0%)	0 (0%)		67 (8.2%)	45 (6.8%)	22 (13%)	
Clinical outcomes								
Symptomatic ICH	0 (0%)	0 (0%)	0 (0%)		18 (2.2%)	0 (0%)	18 (11%)	<0.001
Early neurological improvement	177 (96%)	120 (99%)	57 (89%)	0.005	726 (88%)	607 (92%)	119 (72%)	<0.001
Early neurological deterioration	9 (4.9%)	1 (0.8%)	8 (13%)	0.002	112 (14%)	55 (8.4%)	57 (35%)	<0.001
90-d mRS score	1 (1–3)	1 (0–1)	4 (3–6)	<0.001	2 (1–3)	2 (1–3)	3 (1–5)	<0.001
90-d functional independence, mRS score, 0–2	121 (65%)	121 (100%)	0 (0%)	<0.001	497 (60%)	434 (66%)	63 (38%)	<0.001
90-d bedridden state or death, mRS score, 5–6	27 (15%)	0 (0%)	27 (42%)	<0.001	133 (16%)	90 (14%)	43 (26%)	<0.001
90-d excellent outcomes, mRS score, 0–1	93 (50%)	93 (77%)	0 (0%)	<0.001	344 (42%)	301 (46%)	43 (26%)	<0.001

EVT indicates endovascular thrombectomy; ICH, intracranial hemorrhage; IVT, intravenous thrombolysis; MM, medical management; and mRS, modified Rankin Scale. *P values were calculated using the χ^2 tests for categorical data and the Mann-Whitney *U* tests for continuous or ordinal data.

nonlinear relationship between age and its association with procedural failure in the EVT group ($P=0.49$). Based on these effect sizes, -1 point was assigned to patients aged ≥ 40 years in 15-year increments (eg, a 42-year old would receive -1 point and a 57-year old would receive -2 points), -2 points were assigned to patients without hypertension, and -2 points were assigned to patients without atrial fibrillation (Table 2).

Based on the above regression results, the MAD-MT score was calculated by summation of weighted points, which yields a total of -8 to 49 points (Table 2).

Assessing the Performance of the MAD-MT Score

First, we explored whether the MAD-MT score is associated with significant heterogeneity in treatment effect. Here, in interaction analyses, the MAD-MT score significantly modulated EVT's effectiveness in the total cohort in terms of 90-day mRS score ($P_{\text{interaction}}=0.048$), 90-day functional independence ($P_{\text{interaction}}=0.032$; Figure 2A), and 90-day bedridden state/death ($P_{\text{interaction}}=0.024$; Figure 2B). Sensitivity analyses using the unrounded sum of correlation coefficients from the clinical need and procedural failure models confirmed similar trends ($P_{\text{interaction}}=0.054$, 0.039 , and 0.022 for overall 90-day mRS score, functional independence, and bedridden state/death, respectively). Visual inspection of the marginal probability plots of 90-day functional independence and bedridden state/death shows that an MAD-MT score threshold of 15 may be used to dichotomize patients into those likely to benefit from EVT and those likely to be harmed by EVT (Figure 2A and 2B).

In sensitivity analyses stratified by M2 versus non-M2 occlusion patients, a diverging pattern appeared. Among

M2 occlusion patients, while MAD-MT did not significantly modulate overall mRS score or odds of bedridden state or death ($P_{\text{interaction}}=0.081$ and 0.38 , respectively), higher MAD-MT was significantly associated with a larger treatment benefit of EVT in terms of functional independence ($P_{\text{interaction}}=0.042$). In contrast, among non-M2 occlusion patients, MAD-MT did not modulate overall mRS score or odds of functional independence ($P_{\text{interaction}}=0.039$ and 0.90 , respectively), while higher values were associated with an attenuated harm signal in terms of bedridden state or death ($P_{\text{interaction}}=0.021$). While these observations were intriguing, 3-way interaction analyses did not reveal significant heterogeneity between M2 and non-M2 patients in terms of MAD-MT's ability to characterize treatment effect heterogeneity for overall mRS score, functional independence, or bedridden state or death (3-way $P_{\text{interaction}}=0.55$, 0.19 , and 0.19 , respectively).

Among high MAD-MT score patients (≥ 15 ; $n=293$), compared with MM, EVT was significantly associated with better 90-day outcomes (median mRS score, 3 versus 4 ; $P=0.009$), higher rate of functional independence (47% versus 26% ; $P=0.021$), lower rate of bedridden state/death (25% versus 43% ; $P=0.024$; Figure 3; Table 3), and similar rates of 90-day excellent outcomes (25% versus 19% ; $P=0.48$), early neurologic improvement (64% versus 55% ; $P=0.32$), early neurologic deterioration (12.0% versus 7.1% ; $P=0.52$), and symptomatic ICH (0.4% versus 0.0% ; $P=0.60$; Table 3).

In contrast, among low MAD-MT score patients (<15 ; $n=710$), compared with MM, EVT was significantly associated with worse 90-day outcomes (median mRS score, 2 versus 1 ; $P=0.014$), lower rate of functional independence (66% versus 77% ; $P=0.026$), lower rate of excellent outcome (49% versus 59% ; $P=0.046$), and higher rate of END (14% versus 4.3% ; $P=0.002$). There

Table 2. Predictors of Clinical Need, Procedural Failure, and the Derivation of MAD-MT Score

Component	OR (95% CI); P value	Log(OR)	Score	Interpretation
Outcome: poor outcomes after MM				
NIHSS score	1.24 (1.16 to 1.34); <0.001	0.22	+1 per point	Higher NIHSS increases the need.
No IVT treatment	5.33 (2.24 to 14.0); <0.001	1.67	+7	Lack of IVT increases the need.
Outcome: EVT failure or complications				
Age, y				
<40	1.35 (1.07 to 1.70); <0.012 per 15 y	0.30 per 15 y	0	Older age increases EVT risk.
40 to 54			-1	
55 to 69			-2	
70 to 84			-3	
≥ 85			-4	
No hypertension	1.85 (1.20 to 2.78); 0.004	0.61	-2	Hypertension decreases EVT risk.
No atrial fibrillation	1.61 (1.08 to 2.44); 0.024	0.48	-2	Atrial fibrillation decreases EVT risk.
MAD-MT score		0.25 per point	-8 to 49	Higher MAD-MT score augments EVT's net benefit.

Correlation coefficients were derived from 2 separate multivariable regression models, with one model predicting poor outcomes in the MM cohort considering NIHSS and IVT treatment as predictors and a second model predicting EVT failure or complications in the EVT cohort considering patient age, hypertension, and atrial fibrillation as predictors. EVT indicates endovascular thrombectomy; IVT, intravenous thrombolysis; MAD-MT, medium and distal mechanical thrombectomy; MM, medical management; NIHSS, National Institutes of Health Stroke Scale; and OR, odds ratio.

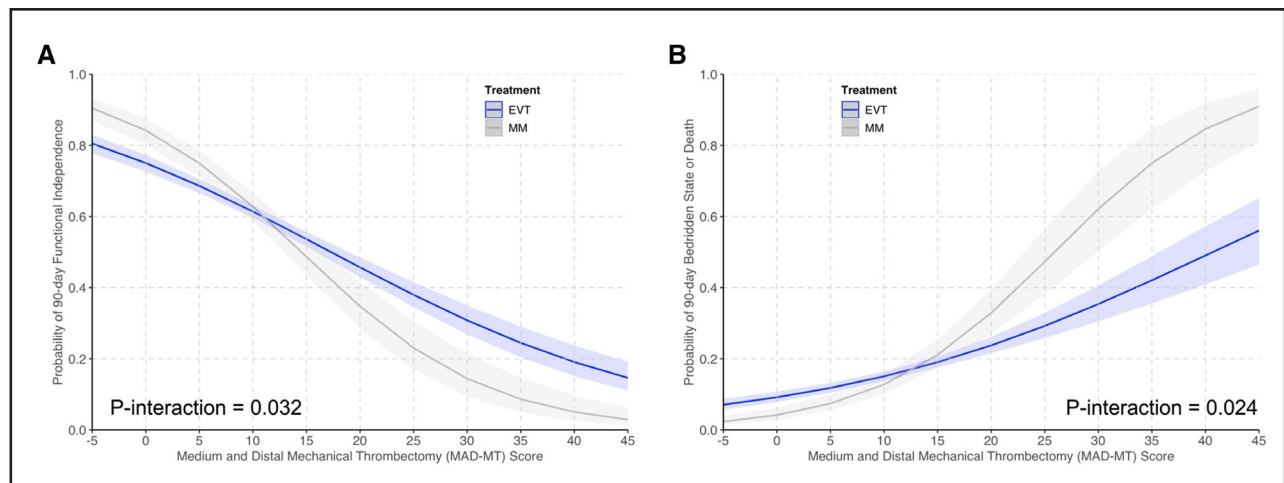


Figure 2. Marginal probability plots of endovascular thrombectomy (EVT) vs medical management (MM) across levels of the medium and distal mechanical thrombectomy (MAD-MT) score.

A. The predicted probabilities of 90-day functional independence. **B.** Predicted probabilities of 90-day bedridden state or death. Shaded areas represent SE. $P_{\text{interaction}}$ values were calculated using multivariable logistic regression models; P values <0.05 are deemed statistically significant.

was also a strong numerical trend among EVT patients toward higher rates of bedridden state/death (12.0% versus 6.4%; $P=0.057$; Figure 3; Table 3) and symptomatic ICH (3.0% versus 0.0%; $P=0.077$; Table 3).

DISCUSSION

In this large, multicenter, multinational retrospective study of 1007 patients with DMVO stroke, we used a multivariable, predictive, risk-based approach to develop the MAD-MT score for guiding patient selection for EVT by integrating both clinical need (predicted poor outcome with MM alone) and procedural risk (likelihood of EVT-related complications or failure). Regression analyses confirmed that the MAD-MT score strongly influenced the treatment effect of EVT on 90-day functional outcomes. Patients with high MAD-MT score (≥ 15) derived substantial benefit from EVT, exhibiting significantly higher odds of functional independence, lower odds of bedridden state or death, and lower overall 90-day mRS score.

Conversely, in patients with low MAD-MT scores (<15), EVT was associated with worse functional outcomes, including reduced odds of independence and a concerning numerical trend toward increased symptomatic hemorrhage and severe disability/death. These findings demonstrate that the MAD-MT score effectively stratifies patients into subgroups where EVT is either significantly beneficial or potentially harmful, supporting its potential role in optimizing treatment decisions for DMVO strokes.

Within the MM cohort, higher NIHSS scores and lack of IVT emerged as key predictors of poor outcomes and clinical need for aggressive treatments, highlighting patients who may derive the greatest benefit from EVT, consistent with prior data.¹⁹ The association between NIHSS score and clinical need is intuitive, as more severe deficits likely reflect a larger salvageable penumbra and, thus, a greater potential gain from reperfusion. Similarly, the absence of IVT may identify patients at higher risk of poor outcomes due to later onset of presentation or lower spontaneous recanalization rates, particularly relevant in

Table 3. EVT Versus MM Outcomes Stratified by the MAD-MT Score

Characteristic, median (Q1–Q3) or n (%)	High MAD-MT score (≥ 15)			Low MAD-MT score (<15)		
	MM (n=42)	EVT (n=251)	P value*	MM (n=141)	EVT (n=569)	P value*
90-d outcomes						
Overall mRS score (shift analysis)	4 (2–6)	3 (1–4)	0.009	1 (0–2)	2 (1–3)	0.014
Functional independence (mRS score, 0–2)	11 (26%)	117 (47%)	0.021	108 (77%)	378 (66%)	0.026
Bedridden state or death (mRS score, 5–6)	18 (43%)	62 (25%)	0.024	9 (6.4%)	71 (12%)	0.057
Excellent outcomes (mRS score, 0–1)	8 (19%)	64 (25%)	0.48	83 (59%)	279 (49%)	0.046
Symptomatic ICH	0 (0%)	1 (0.4%)	1.00	0 (0%)	17 (3.0%)	0.077
Early neurological improvement	23 (55%)	161 (64%)	0.32	48 (34%)	224 (39%)	0.29
Early neurological deterioration	3 (7.1%)	30 (12%)	0.52	6 (4.3%)	82 (14%)	0.002

EVT indicates endovascular thrombectomy; ICH, intracranial hemorrhage; MAD-MT, medium and distal mechanical thrombectomy; and MM, medical management.

* P values were calculated using the χ^2 tests for categorical data and the Mann-Whitney U tests for continuous or ordinal data.

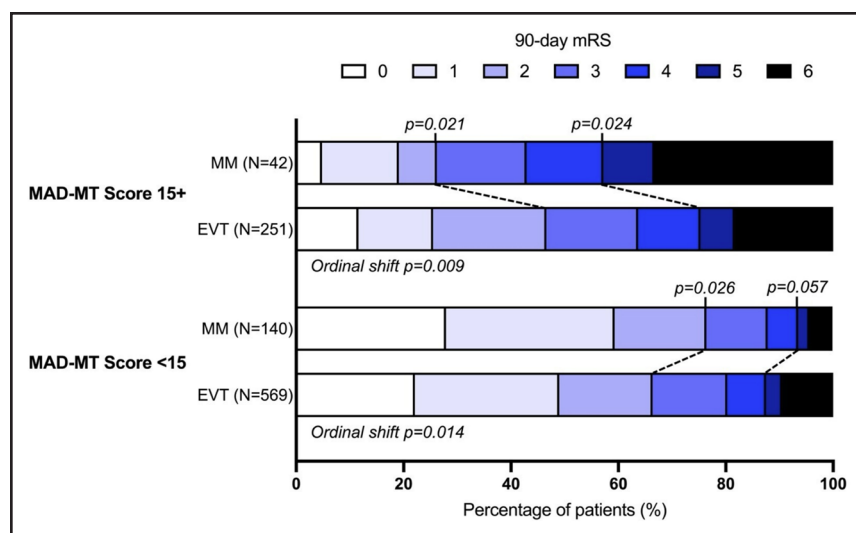


Figure 3. 90-day modified Rankin Scale (mRS) outcomes stratified by treatment modality (medical management [MM] vs endovascular thrombectomy [EVT]) and high vs low medium and distal mechanical thrombectomy (MAD-MT) score.

Rates of functional independence (mRS score, 0–2) and bedridden state or death (mRS score, 5–6) were compared using the χ^2 tests; overall ordinal shifts of 90-day mRS score were assessed using the Wilcoxon rank-sum tests; *P* values <0.05 were deemed statistically significant.

DMVO strokes, where IVT recanalization rates can exceed 50%.^{20–22} While IVT recanalization is not guaranteed, subjecting patients who may recanalize without EVT may introduce unnecessary procedural harm.²³ Prior studies have investigated independent predictors of IVT recanalization for the purposes of stroke imaging triage^{20,22}; similar methods may also be used to select patients for DMVO-EVT, pending future investigation.

Among EVT-treated patients, older age was associated with higher odds of procedural failure and complications, possibly due to increased vessel tortuosity, reduced vascular compliance, or heightened susceptibility to hemorrhagic transformation.^{24–26} In contrast, atrial fibrillation and hypertension were linked to lower procedural risk. The protective effect of atrial fibrillation may stem from the RBC-rich composition of cardioembolic clots, facilitating easier retrieval, while the absence of underlying atherosclerotic disease in these patients may reduce the risk of vascular injury.^{27,28} Hypertension, paradoxically, may confer protection by sustaining collateral flow, thereby preserving the penumbra, slowing the growth of infarct cores.²⁹ However, other reports have suggested that chronic hypertension may be associated with poorer collateral status.³⁰ Thus, our findings that hypertension may be protective require further prospective validation.

Of note, a key insight from our regression analyses to identify predictors of clinical need and procedural complications was that the site of vascular occlusion did not emerge as a significant factor. This suggests that the heterogeneity in DMVO stroke outcomes is driven more by patient-specific clinical and pathophysiological factors (eg, NIHSS score, age, and stroke cause) than by anatomic occlusion site alone. This challenges the conventional emphasis on vessel-level selection criteria and supports a more nuanced, patient-centered approach to EVT decision-making.

Overall, our study introduces a practical, risk-stratified framework to optimize EVT selection in DMVO strokes,

addressing a critical gap in current evidence. While prior randomized controlled trials reported neutral overall effects for DMVO-EVT,^{1,2} our data suggest that treatment effect heterogeneity exists and can be leveraged to maximize benefit while minimizing harm. Prospective validation of the MAD-MT score in randomized trials (eg, as an enrichment tool or stratified randomization factor) or other medium vessel occlusion observational cohorts is warranted. In addition, the MAD-MT score incorporates only clinical factors without consideration of advanced radiographic markers, such as perfusion metrics or collateral status; investigations into these other radiographic markers and future incorporation into the MAD-MT score may further refine risk prediction.^{31–34}

Limitations

Our study has several limitations. First, as a retrospective study, our findings are subject to unmeasured confounding, despite our multivariable approach. Second, procedural techniques and operator experience were not standardized, which may influence outcomes. However, as our study involved 37 participating sites, each with its own practice patterns that may have changed and evolved over time, our results are unlikely to be strongly biased by any particular practice pattern, and our multicenter and international study design likely strengthens the generalizability of our findings. Third, MM cases were underrepresented, which may have contributed to the relatively small sample size of the MM cohort; furthermore, as our study contained mostly M2 occlusion strokes and EVT is generally considered standard of care for these patients before the recent publication of neutral DMVO-EVT trials, it is possible that the MM cohort was enriched with patients deemed unlikely to benefit from EVT treatment. However, as the goal of our study was to identify predictors of poor outcomes among MM patients, this underascertainment bias is not expected to have a major impact

on the validity of our findings. Of note, due to the smaller sample size in the MM cohort, the event-per-variable for our regression model was low at 4.9; this may have led to model overfitting despite backward variable selection though NIHSS score and IVT use are well-recognized predictors of outcomes for patients with stroke managed with MM alone, so our model specifications remain biologically plausible. Overfitting was less of a concern for the model predicting procedural failure in the EVT cohort (event-per-variable, 12.7). Finally, as the MAD-MT score was derived from a retrospective study, its use requires external validation in a prospective, randomized controlled setting or in a similar DMVO cohort. Of note, while single-study development of predictive models is at risk of statistical overfitting and limited generalizability, our predictions (clinical need and procedural risk) and our eventual outcome of interest (treatment response to EVT) are distinct, thus limiting the risk of overfitting in our study.

Conclusions

The MAD-MT score provides a clinically actionable tool to identify patients with DMVO stroke most likely to benefit from EVT while avoiding harm in those unlikely to derive meaningful advantage. By integrating both clinical need and procedural risk, this approach can potentially aid patient selection in clinical practice for EVT versus MM, future trial design, and the implementation of personalized stroke care in this challenging population. External validation is now needed to confirm the clinical utility and reliability of the MAD-MT score.

ARTICLE INFORMATION

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The podcast and transcript are available at <https://www.ahajournals.org/str/podcast>.

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Supplemental Material

Collaborators: MAD-MT Investigators
STROBE Checklist

REFERENCES

1. Psychogios M, Brehm A, Ribo M, Rizzo F, Strbian D, Rätty S, Arenillas JF, Martínez-Galdámez M, Hajdu SD, Michel P, et al. Endovascular treatment for stroke due to occlusion of medium or distal vessels. *N Engl J Med*. 2025;392:1374–1384. doi: 10.1056/nejmoa2408954
2. Goyal M, Ospel JM, Ganesh A, Dowlatshahi D, Volders D, Möhlenbruch MA, Jumaa MA, Nimjee SM, Booth TC, Buck BH, et al. Endovascular treatment of stroke due to medium-vessel occlusion. *N Engl J Med*. 2025;392:1385–1395. doi: 10.1056/nejmoa2411668
3. Dmytriw AA, Musmar B, Salim H, Ghazy S, Siegler JE, Kobeissi H, Shaikh H, Khalife J, Abdalkader M, Klein P, et al; MAD-MT Consortium. Incidence and clinical outcomes of perforations during mechanical thrombectomy

- for medium vessel occlusion in acute ischemic stroke: a retrospective, multicenter, and multinational study. *Eur Stroke J*. 2024;9:328–337. doi: 10.1177/23969873231219412
4. Salim HA, Yedavalli V, Musmar B, Adeeb N, El Naamani K, Henninger N, Sundararajan SH, Kühn AL, Khalife J, Ghozy S, et al. Endovascular therapy versus best medical management in distal medium middle cerebral artery acute ischaemic stroke: a multinational multicentre propensity score-matched study. *J Neurol Neurosurg Psychiatry*. 2024;96:239. doi: 10.1136/jnnp-2024-333669
 5. Chen H, Colasurdo M, Khunte M, Malhotra A, Gandhi D. Post-thrombectomy subarachnoid hemorrhage: incidence, predictors, clinical relevance, and effect modulators. *Diagnostics (Basel)*. 2024;14:1856. doi: 10.3390/diagnostics14171856
 6. Chen H, Khunte M, Colasurdo M, Malhotra A, Gandhi D. Thrombectomy vs medical management for posterior cerebral artery stroke: systematic review, meta-analysis, and real-world data. *Neurology*. 2024;102:e209315. doi: 10.1212/WNL.0000000000209315
 7. Chen H, Khunte M, Malhotra A, Gandhi D, Colasurdo M. Endovascular thrombectomy versus medical management for moderate-to-severe anterior cerebral artery occlusion stroke. *J Neurol*. 2024;271:6247–6254. doi: 10.1007/s00415-024-12582-z
 8. Chen H, Colasurdo M, Phipps MS, Miller TR, Cherian J, Marino J, Cronin CA, Wozniak MA, Gandhi D, Chaturvedi S, et al. The BAND score: a simple model for upfront prediction of poor outcomes despite successful stroke thrombectomy. *J Stroke Cerebrovasc Dis*. 2024;33:107608. doi: 10.1016/j.jstrokecerebrovasdis.2024.107608
 9. Amin SM, Chen H, Rewinkel SE, Lockwood DA, Kim D, Priest RA, Nesbit GD, Liu JJ, Horikawa M, Clark WM, et al. The PANDA score: external validation and modification of a simple upfront prediction tool for poor outcomes despite successful stroke thrombectomy. *Clin Neurol Neurosurg*. 2025;249:108722. doi: 10.1016/j.clineuro.2025.108722
 10. Siegler JE, Koneru M, Qureshi MM, Doheim M, Nogueira RG, Martinez-Majander N, Nagel S, Penckofer M, Demeestere J, Puetz V, et al. CLEAR thrombectomy score: an index to estimate the probability of good functional outcome with or without endovascular treatment in the late window for anterior circulation occlusion. *J Am Heart Assoc*. 2024;13:e034948. doi: 10.1161/jaha.124.034948
 11. Kremers F, Venema E, Duvekot M, Yo L, Bokkers R, Lycklama A, Nijeholt G, van Es A, van der Lugt A, Majoie C, et al. Outcome prediction models for endovascular treatment of ischemic stroke: systematic review and external validation. *Stroke*. 2022;53:825–836. doi: 10.1161/STROKEAHA.120.033445
 12. Yedavalli V, Salim H, Musmar B, Adeeb N, El Naamani K, Henninger N, Sundararajan SH, Kühn AL, Khalife J, Ghozy S, et al. Pretreatment predictors of very poor clinical outcomes in medium vessel occlusion stroke patients treated with mechanical thrombectomy. *Int J Stroke*. 2024;19:1123–1133. doi: 10.1177/17474930241270524
 13. van Swieten JC, Koudstaal PJ, Visser MC, Schouten HJ, van Gijn J. Interobserver agreement for the assessment of handicap in stroke patients. *Stroke*. 1988;19:604–607. doi: 10.1161/01.str.19.5.604
 14. Zaidat OO, Yoo AJ, Khatri P, Tomsick TA, von Kummer R, Saver JL, Marks MP, Prabhakaran S, Kallmes DF, Fitzsimmons B-FM, et al; Cerebral Angiographic Revascularization Grading (CARG) Collaborators. Recommendations on angiographic revascularization grading standards for acute ischemic stroke. *Stroke*. 2013;44:2650–2663. doi: 10.1161/STROKEAHA.113.001972
 15. Hacke W, Kaste M, Fieschi C, von Kummer R, Davalos A, Meier D, Larrue V, Bluhmki E, Davis S, Donnan G, et al. Randomised double-blind placebo-controlled trial of thrombolytic therapy with intravenous alteplase in acute ischaemic stroke (ECASS II). *Lancet*. 1998;352:1245–1251. doi: 10.1016/s0140-6736(98)08020-9
 16. Barber PA, Demchuk AM, Zhang J, Buchan AM. Validity and reliability of a quantitative computed tomography score in predicting outcome of hyperacute stroke before thrombolytic therapy. ASPECTS Study Group. Alberta Stroke Programme Early CT Score. *Lancet*. 2000;355:1670–1674. doi: 10.1016/s0140-6736(00)02237-6
 17. Rothwell P, Warlow C. Prediction of benefit from carotid endarterectomy in individual patients: a risk-modelling study. *Lancet*. 1999;353:2105–2110. doi: 10.1016/s0140-6736(98)11415-0
 18. Portet S. A primer on model selection using the Akaike information criterion. *Infect Dis Model*. 2020;5:111–128. doi: 10.1016/j.idm.2019.12.010
 19. Kühn AL, Puri AS, Salim HA, Musmar B, Ghozy S, Siegler J, Shaikh H, Khalife J, Abdalkader M, Klein P, et al; MAD-MT Consortium. Multicenter evaluation of mechanical thrombectomy for distal medium vessel occlusions with National Institute of Health Stroke Scale scores ≥ 6 and ≤ 6 . *J Neurol*. 2024;271:5853–5863. doi: 10.1007/s00415-024-12537-4
 20. Colasurdo M, Chen H, Schrier C, Khalid M, Khunte M, Miller TR, Cherian J, Malhotra A, Gandhi D. Predictors for large vessel recanalization before stroke thrombectomy: the HALT score. *J Neurointerv Surg*. 2023;16:237–242. doi: 10.1136/jnis-2023-020220
 21. Seners P, Turc G, Maier B, Mas J-L, Oppenheim C, Baron J-C. Incidence and predictors of early recanalization after intravenous thrombolysis. *Stroke*. 2016;47:2409–2412. doi: 10.1161/STROKEAHA.116.014181
 22. Chen H, Skorseth P, Rewinkel S, Kim D, Amin S, Shakal S, Priest R, Nesbit G, Clark W, Colasurdo M. Development of a machine learning model to predict changes in neuroimaging profiles among acute ischemic stroke patients following delayed transfer for endovascular thrombectomy. *Neuroradiology*. 2025;67:1827–1836. doi: 10.1007/s00234-025-03600-6
 23. Faizy TD, Yedavalli V, Salim HA, Lakhani DA, Musmar B, Adeeb N, Essibayi MA, Daraghma M, El Naamani K, Henninger N, et al. Clinical outcomes of patients with unsuccessful mechanical thrombectomy versus best medical management of medium vessel occlusion stroke in the middle cerebral artery territory. *J Neurointerv Surg*. 2025;18:155–165. doi: 10.1136/jnis-2024-022642
 24. Chen H, Jindal G, Miller TR, Gandhi D, Chaturvedi S. Stroke thrombectomy in the elderly: efficacy, safety, and special considerations. *Stroke Vasc Interv Neurol*. 2023;3. doi: 10.1161/svin.122.000634
 25. Chen H, Colasurdo M, Khunte M, Malhotra A, Gandhi D. Endovascular thrombectomy versus medical management for acute basilar artery occlusion stroke in the elderly. *Am J Neuroradiol*. 2024;46:278–284. doi: 10.3174/ajnr.a8473
 26. Chen H, Khunte M, Colasurdo M, Malhotra A, Gandhi D. Intravenous thrombolysis prior to endovascular thrombectomy in elderly stroke patients: an analysis of the National Inpatient Sample database. *J Neurol Sci*. 2023;454:120842. doi: 10.1016/j.jns.2023.120842
 27. Pillai P, Bush SJ, Kusuma Y, Churilov L, Dowling RJ, Luu VD, Davis SM, Mitchell PJ, Yan B. Atrial fibrillation is associated with higher first pass effect following thrombectomy for large vessel occlusion. *J Neurointerv Surg*. 2024;16:652–656. doi: 10.1136/jnis-2023-020512
 28. Yedavalli V, Salim HA, Musmar B, Adeeb N, Naamani KE, Henninger N, Sundararajan SH, Kühn AL, Khalife J, Ghozy S, et al; MAD MT Investigators. Predictors and outcomes of excellent recanalization versus successful recanalization after thrombectomy in proximal and distal medium vessel occlusion strokes: a multinational study. *Stroke Vasc Interv Neurol*. 2024;4:e001421. doi: 10.1161/SVIN.124.001421
 29. Jiang B, Churilov L, Kanesan L, Dowling R, Mitchell P, Dong Q, Davis S, Yan B. Blood pressure may be associated with arterial collateralization in anterior circulation ischemic stroke before acute reperfusion therapy. *J Stroke*. 2017;19:222–228. doi: 10.5853/jos.2016.01739
 30. Fujita K, Tanaka K, Yamagami H, Ide T, Ishiyama H, Sonoda K, Satow T, Takahashi JC, Ihara M, Koga M, et al. Detrimental effect of chronic hypertension on leptomeningeal collateral flow in acute ischemic stroke. *Stroke*. 2019;50:1751–1757. doi: 10.1161/STROKEAHA.119.025142
 31. Salim H, Lakhani DA, Balar A, Musmar B, Adeeb N, Hoseinyazdi M, Luna L, Deng F, Hyson NZ, Mei J, et al. Follow-up infarct volume on fluid attenuated inversion recovery (FLAIR) imaging in distal medium vessel occlusions: the role of cerebral blood volume index. *J Neurol*. 2024;271:3389–3397. doi: 10.1007/s00415-024-12279-3
 32. Salim HA, Vagal V, Lakhani DA, Mei J, Luna L, Aziz Y, Balar A, Dmytriw AA, Guenego A, Musmar B, et al. Association of pretreatment perfusion imaging parameters with 90-day excellent functional outcomes in anterior circulation distal medium vessel occlusion stroke. *AJNR Am J Neuroradiol*. 2025;46:892–899. doi: 10.3174/ajnr.A8584
 33. Koneru M, Lakhani DA, Xu R, Salim HA, Urrutia VC, Marsh EB, Dmytriw AA, Guenego A, Llinas RH, Hillis AE, et al. Cerebral blood volume index in the era of thrombectomy-treated large and medium vessel ischemic strokes. *J Neurointerv Surg*. 2025;17:1289–1293. doi: 10.1136/jnis-2024-022609
 34. Maguida G, Shuaib A. Collateral circulation in ischemic stroke: an updated review. *J Stroke*. 2023;25:179–198. doi: 10.5853/jos.2022.02936